

Exploratory Data Analysis (EDA) Report

Project: AI Credit Risk Prediction
Dataset: Loan Default Dataset (Loan_default.csv)
Milestone: Milestone 2 – Data Ingestion & Initial EDA

1. Introduction

Exploratory Data Analysis (EDA) is the process of examining and understanding a dataset before applying machine learning models. It helps identify the structure, patterns, relationships, and potential issues within the data.

The goal of EDA in this project is to analyze the **Loan Default dataset** to understand the features affecting loan repayment behavior and prepare the dataset for machine learning model development.

2. Dataset Description

The dataset used in this project is **Loan_default.csv**, which contains information about loan applicants and their financial characteristics. The dataset is used to predict whether a customer will **default on a loan**.

Each row represents a **loan application**, and each column represents a **feature describing the applicant's financial or personal attributes**.

Dataset Columns

Column	Description
LoanID	Unique identifier for each loan record
Age	Age of the applicant
Income	Annual income of the applicant
LoanAmount	Amount of loan taken
CreditScore	Credit score representing creditworthiness
MonthsEmployed	Number of months the applicant has been employed
NumCreditLines	Number of active credit lines
InterestRate	Interest rate applied to the loan
LoanTerm	Duration of the loan in months
DTIRatio	Debt-to-Income ratio of the applicant
Education	Education level of the applicant

Column	Description
EmploymentType	Type of employment
MaritalStatus	Marital status of the applicant
HasMortgage	Indicates if the applicant has a mortgage
HasDependents	Indicates if the applicant has dependents
LoanPurpose	Purpose of the loan
HasCoSigner	Indicates whether the loan has a co-signer
Default	Target variable indicating loan repayment status

3. Data Ingestion

The dataset was ingested into a relational database using **PostgreSQL**.

A Python script (load_dataset.py) was used to perform the ingestion process using **pandas** and **SQLAlchemy**.

Steps involved in data ingestion:

1. Load the dataset from the CSV file.
2. Establish a connection to the PostgreSQL database.
3. Insert the dataset into a database table named **loan_default**.

This ensures the dataset is stored in a **structured database format**, allowing efficient querying and management of data.

4. Sample Data Record

A sample record from the dataset is shown below:

LoanID	Age	Income	LoanAmount	CreditScore	Default
128027	56	85994	50587	520	0

Interpretation of this record:

- The applicant is **56 years old**.
- The applicant has an **annual income of 85,994**.
- The loan amount requested is **50,587**.
- The credit score is **520**, indicating moderate creditworthiness.
- The **Default value is 0**, meaning the loan was successfully repaid.

This example illustrates how the dataset captures financial and demographic information used to predict loan defaults.

5. Data Preprocessing

Before building machine learning models, the dataset was cleaned and prepared.

5.1 Handling Missing Values

Missing values were identified using:

```
df.isnull().sum()
```

Rows containing missing values were removed to ensure the dataset contained complete records.

5.2 Encoding Categorical Variables

Several columns contained categorical data, such as:

- Education
- EmploymentType
- MaritalStatus
- LoanPurpose

These columns were converted into numeric values using encoding techniques so they could be processed by machine learning algorithms.

5.3 Removing Unnecessary Columns

The **LoanID** column was removed before training the model because it is only an identifier and does not contribute to prediction.

6. Statistical Summary

Statistical analysis was performed using:

```
df.describe()
```

This provided insights into the dataset, including:

- Mean values
- Minimum and maximum values
- Standard deviation

Key observations include:

- Income levels vary widely among applicants.

- Loan amounts differ based on applicant financial profiles.
- Credit scores provide an indication of borrower reliability.

These statistics help understand the **distribution of numerical features**.

7. Target Variable Analysis

The **Default column** represents the target variable used for prediction.

Value Meaning

- | | |
|---|--------------------------|
| 0 | Loan repaid successfully |
| 1 | Loan defaulted |

Analyzing this variable helps determine how many applicants repaid their loans versus those who defaulted. This information is important for training accurate machine learning models.

8. Conclusion

The Exploratory Data Analysis (EDA) helped understand the structure and characteristics of the loan dataset.

Key outcomes include:

- The dataset was successfully ingested into **PostgreSQL**.
- Data cleaning and preprocessing ensured that the dataset contained complete and usable records.
- Categorical variables were encoded into numeric values.
- Statistical analysis provided insights into applicant financial characteristics.

After completing EDA and preprocessing, the dataset is now **ready for machine learning model training and prediction tasks in the next stages of the project**.